

PROJECT BRIEF · GUINEA

Nimba Lumière

Smart Solar Community Totem

Off-grid light that protects, informs, and brings people together.

Lighting

Main streetlight + 4 community boosters

Warning

5-colour weather beacon (flag-inspired)

Charging

USB port with mechanical paddle timer



Two problems, one streetlight

In many Guinean communities, the grid is unreliable and weather warnings rarely reach people in time. Guinea Lumière addresses both with a single, off-grid object placed where people already gather.

The dark street

Households rely on torches, candles, and oil lamps after sunset. Streets without light see drops in safety, trade, and informal social life. Power cuts leave students and workers without options.

The unseen storm

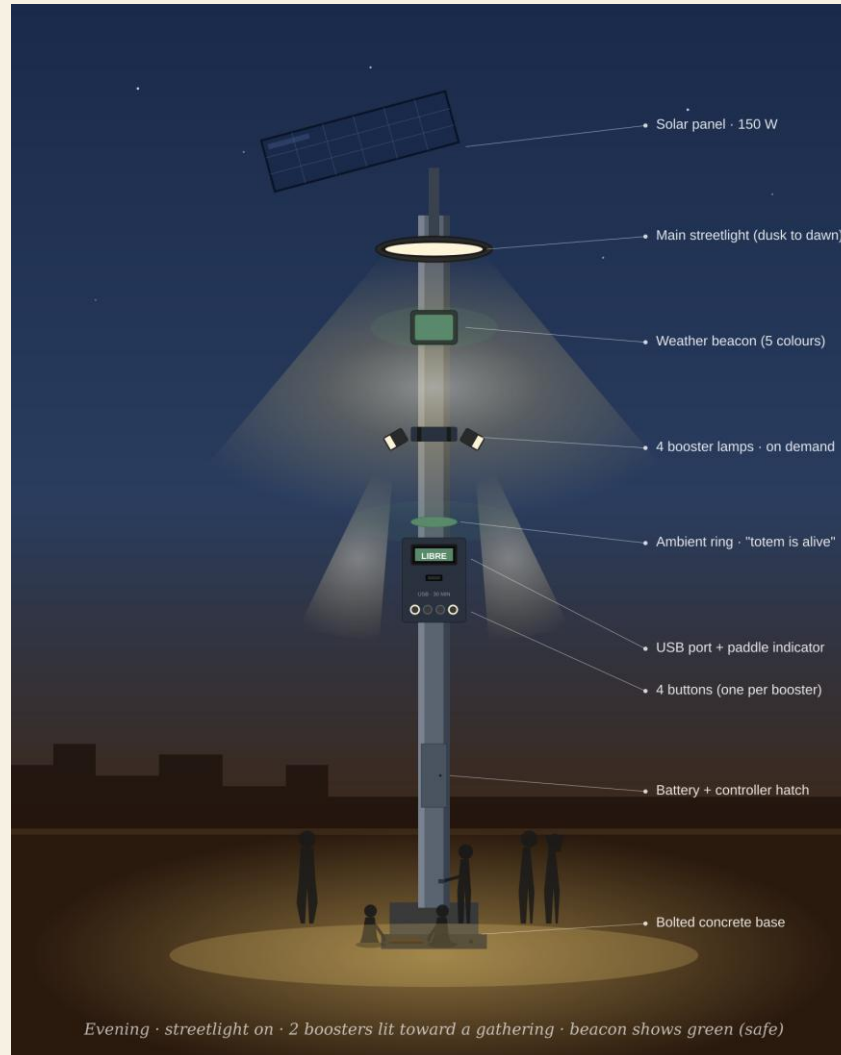
Recent floods in Conakry and Siguiri have displaced families and damaged services. Many people do not receive timely, visual warnings before heavy rain — they react after, not before.

THE RESPONSE

A fully solar, off-grid streetlight that gives shared light, signals weather risk in the colours of the Guinean flag, and turns a single point of illumination into a place for study, trade, and community life.

What the totem looks like in use

Evening scene: main streetlight on automatically. Two boosters lit by community choice toward a gathering.



DESIGN HIGHLIGHTS

Streetlight first, boosters on demand

A main lamp on top lights the street automatically from dusk to dawn. Four smaller booster lamps stay off until the community switches them on for an activity.

The beacon speaks the flag

Weather warnings cycle through green, yellow, red and blue - the colours of Guinea, plus blue for water. The totem speaks a visual language people already know.

Made for gathering

The booster lamps create a warm pool of light only when needed, for study, games, hair braiding, trade, meetings. Battery is saved on quiet nights.

Locally serviceable

Hatch at the base gives technicians access to the battery, controller, and SIM. The pole is bolted, not buried — it can be moved if the community moves.

The totem speaks the flag

The colours of the weather beacon are not arbitrary. They are the colours of Guinea — each one carrying meaning the community already holds.



VERT / GREEN

Our forests, our vegetation

All is well. Steady glow on calm nights.



JAUNE / YELLOW

The richness of our soil

Pay attention -something is shifting.



BLEU / BLUE

The sky that brings rain

Heavy rain is coming. Prepare.



ROUGE / RED

The blood of our martyrs — those who acted for others

Now we act. Now we help each other.

*Vert comme nos forêts. Jaune comme notre terre. Rouge comme le sang des nôtres. Bleu comme le ciel qui pleure.
Green like our forests. Yellow like our soil. Red like the blood of our own. Blue like the sky that weeps.*

Four roles in one object

Each role is delivered by a dedicated subsystem on the same pole.

01

Light the street, lift the gathering

A main streetlight comes on automatically at dusk and goes off at dawn. Four booster lamps stay off until someone presses a button each lights one zone around the pole for studying, trade, or community activities.

02

Warn in the colours of the flag

Green at rest, yellow when conditions shift, blue when rain is coming, red when action is needed. Each colour is both a functional warning and a cultural message -the totem speaks a language people already know.

03

Power one phone at a time

A single USB-A port offers timed 30-minute sessions, governed by a mechanical paddle indicator. A single audible click signals the end of the session -no buzzer, no screen, no language barrier.

04

Stand on its own

Fully off-grid: solar panel, lithium iron phosphate battery, and a local 4G SIM for diagnostics. Data is stored locally on an SD card and syncs to cloud only when signal is available.

Components-power system

The energy backbone. Sized for two days of autonomy and Guinea's heat and humidity.

GENERATION	STORAGE	CONDITIONING
Solar panel 150 W (monocrystalline) Converts sunlight to DC. Outperforms poly in diffuse rainy-season light. Tilt-adjustable bracket, anti-theft cage. Mounting bracket Holds the panel at ~10° tilt. Allows one technician to re-aim or remove for maintenance.	LiFePO4 battery 12.8 V / 100 Ah Stores ~1.28 kWh. Lasts 8–10 years, tolerates Guinea's heat, no thermal runaway risk near children. IP65 lockable enclosure Houses the battery inside the pole base, off the ground to survive flooding, locked against tampering.	MPPT charge controller 20 A Extracts maximum power from the panel, manages battery cycles, exposes Bluetooth diagnostics. Low-voltage disconnect (LVD) Sheds non-essential loads before the warning system loses power. Lifesafety stays last to fail. DC-DC converter (12 V → 5 V / 3 A) Steps voltage down for the USB port. Isolated, so a cable short does not affect the rest. Surge protection Protects panel input and load output from lightning and transients during West African storms.

Components - lighting, warning & USB

The visible layer. Main lamp runs automatically; boosters are community-controlled.

LIGHTING

Main streetlight (30 W LED head)

Mounted at the top of the pole. Automatic dusk-to-dawn via light sensor. Wide light cone for street-level illumination.

4 × booster lamps (10 W each)

On a collar below the beacon. Each lights one zone (N, S, E, W). Switched on by button — stay on until manually switched off.

Ambient ring LED (low power)

Soft green glow at mid-pole. The 'totem is alive' indicator. Tells passers-by the system is on, even by day.

4 backlit booster buttons

Chest-height (1.2 m), one per booster. Button backlight glows when its lamp is on, so users see which zones are lit.

WARNING

5-colour RGB warning beacon

Top of the pole. Switches between green (safe), yellow (caution), blue (rain coming), red (act now), white (system on).

Piezo siren 100 dB

Triggered only at red-level alert. Audible across the gathering area. Sits on the priority power rail.

Speaker 5 W

Plays short pre-recorded messages in French and local languages, so warnings are understood even by those who cannot see the beacon.

USB CHARGING

USB-A port (IP54)

Recessed flap to keep dust and rain out. Limited to one device per session for fairness.

Pivoting paddle indicator

Mechanical paddle behind a small window. Green LIBRE when free, red OCCUPÉ during a session. Held by a latch, not by power.

Solenoid + mechanical latch

Brief pulse flips the paddle at plug-in. Second pulse releases the latch at 30 min - paddle falls back, makes a single click.

Components - control, sensing & structure

The intelligence that ties it together, and the body that holds it up.

CONTROL & CONNECTIVITY

Microcontroller (ESP32 or Pi Zero 2 W)

Runs all logic: lamp scheduling, warning escalation, sensor reading, USB sessions, paddle solenoids.

4G LTE modem + local SIM

Sends diagnostics and weather logs to cloud when signal is available. The totem keeps working with or without it.

SD card 32 GB

Local storage for sensor history, USB usage stats, battery state, and event logs. Syncs upward only when 4G is up.

Real-time clock module

Keeps timers and dusk/dawn switching accurate even when offline and after a power cut.

SENSING

Tipping-bucket rain gauge

Measures rainfall intensity. Feeds the warning logic and the long-term weather log.

BME280 pressure + humidity sensor

Pressure drops sharply before tropical storms. The earliest signal the system has.

Light sensor (LDR)

Triggers the automatic dusk-to-dawn switching of the main streetlight.

Tilt / vibration sensor

Anti-theft. A sudden shake or tilt triggers a 4G alert and lights up the beacon.

STRUCTURE

Steel pole 4–5 m, galvanized

Hot-dip galvanizing resists coastal humidity and salt. Hollow core routes wiring safely from base to head.

Bolted base plate

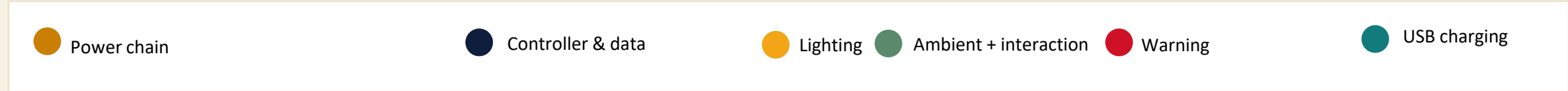
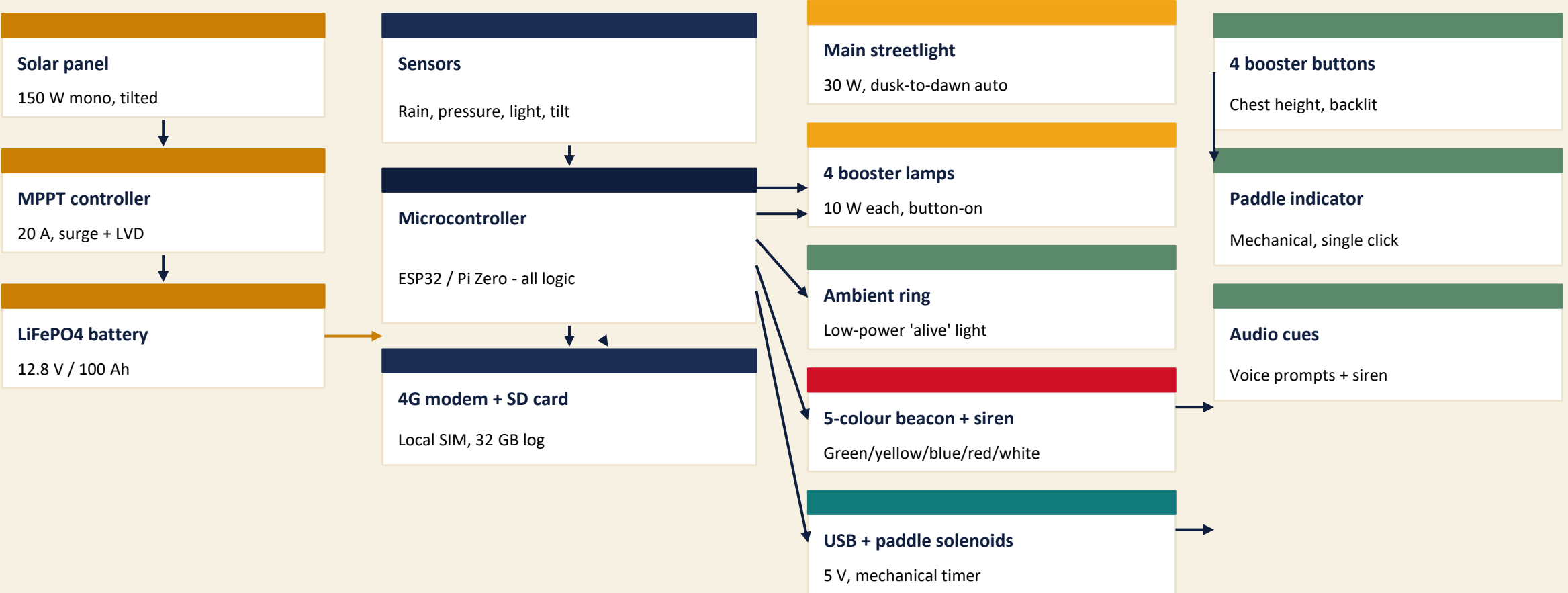
Concrete footing with bolts, not a buried pole. The totem can be unbolted and relocated as a community grows.

Maintenance hatch

Lockable access door at the base for battery, controller, and SIM. Keyed to local technicians only.

System architecture

Power flows down the left. The microcontroller in the middle reads sensors and drives every load.



How it all connects

Reading the architecture as a short story, in five steps.

1

Panel charges battery

The 150 W panel feeds the MPPT controller, which conditions the current and charges the LiFePO4 battery. Surge protection and a low-voltage disconnect sit on this path.

2

Battery feeds the 12 V bus

All loads share one 12 V DC bus from the battery. A DC-DC converter drops 12 V to 5 V for the USB port; lighting, beacon, and solenoids run directly off 12 V.

3

Microcontroller decides

The MCU reads sensors and booster buttons, schedules the main streetlight by dusk/dawn, escalates the beacon colour, and runs the USB timer with the paddle solenoids.

4

Loads respond

The main lamp turns on at dusk automatically. Booster lamps light only when buttons are pressed. The beacon shifts colour as conditions change. Plugging in fires the paddle solenoid.

5

Data logs locally, syncs when it can

Every event - weather, USB usage, battery state, faults - is written to the local SD card. The 4G modem uploads when there is signal. No signal, no problem: the totem keeps working.

Lighting behaviour

Automatic for the street. On-demand for the community.

Main streetlight

Automatic - runs every night

- 30 W LED at the top of the pole
- Turns on automatically at dusk via the light sensor
- Turns off automatically at dawn
- Provides the default illumination of the street
- No button - works the same every night
- Approx. 12 hours of operation per night

4 booster lamps

Community-controlled — on demand

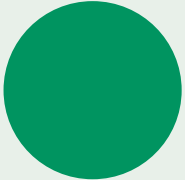
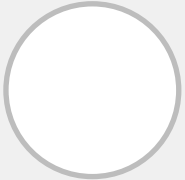
- 10 W each, one per cardinal direction (N, S, E, W)
- Off by default - battery preserved on quiet nights
- Pressed by anyone passing - backlit button per lamp
- Light just the zone where activity is happening
- Stay on until someone presses the button again to switch off
- Auto-off at dawn if anyone forgets

WHY THIS WORKS

Street is always lit - safety guaranteed. Boosters lit only when needed — battery preserved. Community decides where to add light, not engineers.

Weather warning logic

Five colours, three escalation stages, plus a calm-night state and a system state.

 GREEN <i>Our forests, our nature</i> MEANING All is well TRIGGER Calm conditions, no weather risk	 WHITE <i>System on, peace</i> MEANING Normal night, light burning TRIGGER Default at dusk, no warnings	 YELLOW <i>The richness of our soil</i> MEANING Pay attention TRIGGER Pressure shifting, conditions changing	 BLUE <i>The sky that brings rain</i> MEANING Heavy rain is coming TRIGGER Rain detected or forecast within 3 hours	 RED <i>The blood of those who acted for others</i> MEANING Now we act together TRIGGER Flood imminent or in progress + siren + voice
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PRIORITY POWER RULE

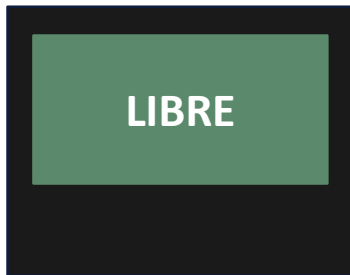
If battery drops below 30 %, USB and boosters shut down first. The beacon, siren, and main streetlight stay powered.

USB charging- the pivoting paddle

A mechanical indicator behind a small window. Almost zero standby power. A single click marks the end of a session.

Before plug-in

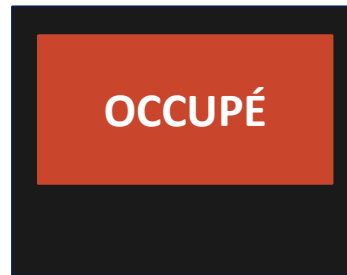
Paddle resting at LIBRE



Gravity holds the paddle in the LIBRE position. Zero power. Anyone can see the port is free from across the street.

While charging

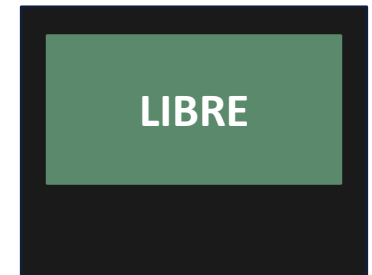
Paddle latched at OCCUPÉ



Plug-in fires a half-second solenoid pulse. A mechanical hook catches and holds the paddle. No more power used during the 30 minutes.

Session over

Paddle drops back, single click



At 30 min, a second pulse releases the hook. The paddle falls under its own weight and strikes a metal stop — the click is the audible alarm.

POWER NOTE

Total power per session: ~0.02 Wh. Roughly 50× less than an LED-ring timer. The paddle holds its state by mechanics, not electricity.

At a glance

The shape of one unit, and what to decide next.

150 W

solar panel

1.28 kWh

LiFePO4 storage

≈ 2 days

autonomy buffer

5

beacon colours

INDICATIVE BILL OF MATERIALS = T B D

NEXT DECISIONS

Pilot site: select one urban and one rural location to test usage, battery cycles, and weather alert response.

Cultural plaque: tbd.

Network choice: 4G carrier

Manufacturing